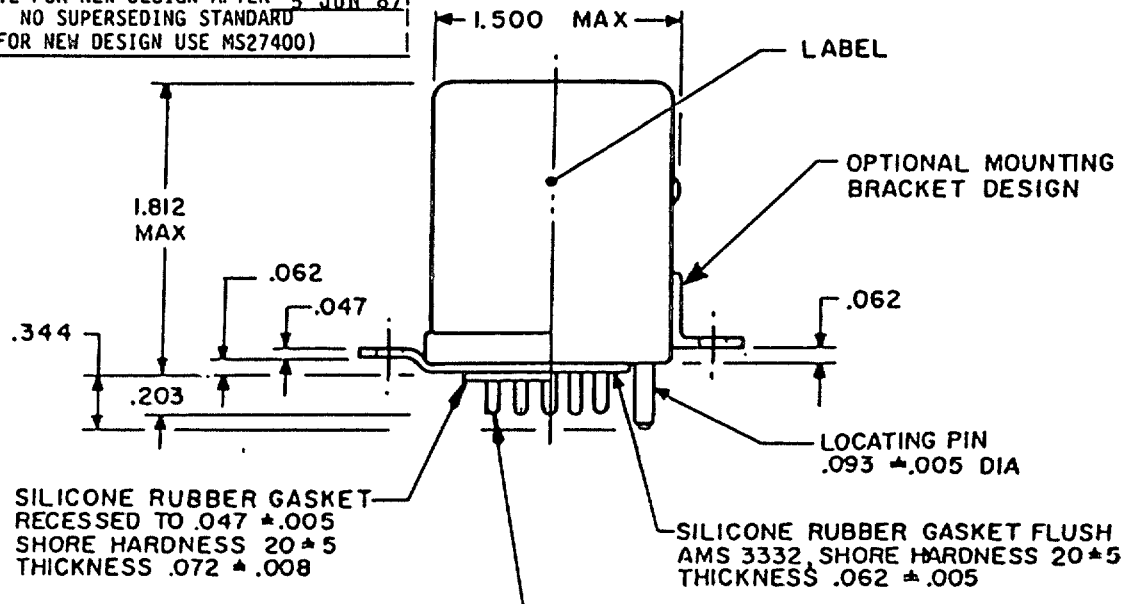
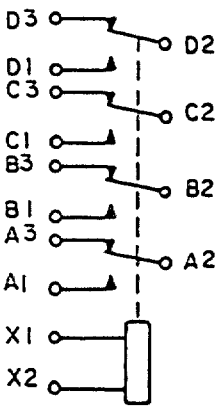
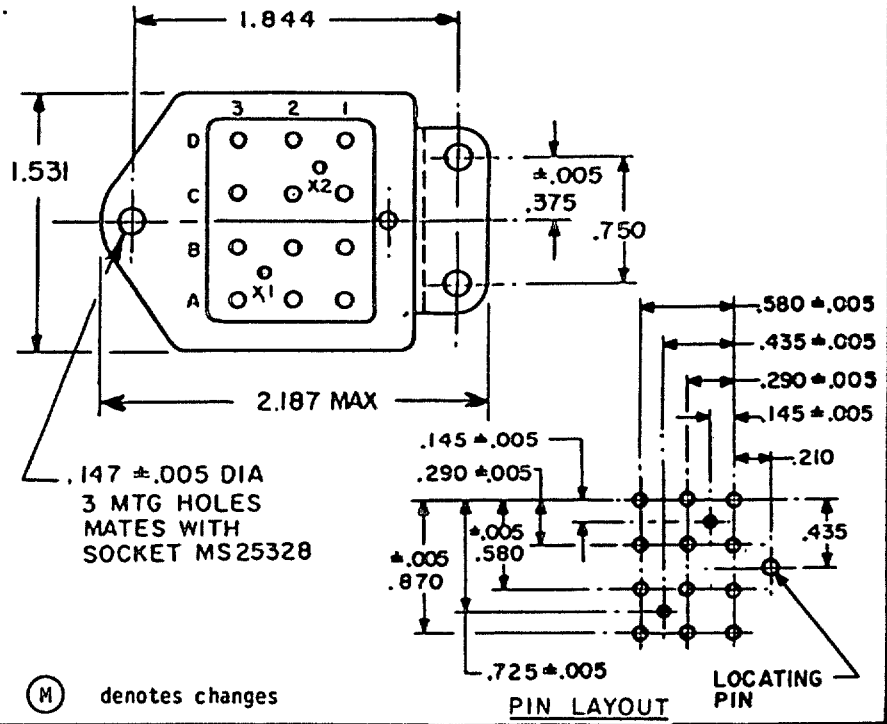
FED. SUP CLASS
5945

INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD
(FOR NEW DESIGN USE MS27400)





CIRCUIT DIAGRAM
(SEE NOTE 4)



Ⓜ denotes changes

APPROVED 12 May 58 REVISED Ⓜ 20 Jan 1989

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99

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P.A	International Interest	TITLE	MILITARY STANDARD
USAF - 85 Other Cust Navy - AS		RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, SOCKET MOUNTED, HERMETICALLY SEALED	MS25327
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

5945-0792-06

AMSC N/A

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Inches	mm	Inches	mm
.001	0.03	.290	7.37
.005	0.13	.344	8.74
.008	0.20	.375	9.52
.040	1.02	.435	11.05
.047	1.19	.580	14.73
.062	1.57	.725	18.42
.0625	1.588	.750	19.05
.072	1.83	.870	22.10
.093	2.36	1.500	38.10
.145	3.68	1.531	38.89
.147	3.73	1.821	46.02
.203	5.16	1.844	46.84
.210	5.33	2.187	55.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. The use of diodes on ac relays is optional. Actual application must be shown on label.
5. Pins to be perpendicular to header surface within one degree.
6. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that positively and permanently identifies each terminal location specified herein.
7. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
8. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS25327-	Type	Coil	Terminal type	Max weight in pounds
D1	1	dc	Plug in	.43
A1	1	ac	Plug in	.45

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Review activities: Army - EC Navy - Air Force - 11, 99
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Review activities: Army - Navy - EC Air Force - 11, 99

User activities: Army - Navy - Air Force -

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Procurement Specification MIL-R-6106

P.A USAF - 85 Other Cust Navy - AS

International Interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, SOCKET MOUNTED, HERMETICALLY SEALED.

MILITARY STANDARD MS25327

PAGE 3 OF 5

FED. SUP CLASS 5945

(M) TABLE II. Operating characteristics.

MS part no. MS25272-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce			
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
01	X1, X2	28	dc	N/A	29	0.350	18	19.5	22.5	7.0	1.5	20	20	2	2		
A1	X1, X2	115	400 5/	N/A	122	0.073	90	95	103	30	5.0	25	50	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25272-A1 may be used on 60 Hz if maximum ambient temperature is limited to 85°C (maximum coil current shall be 0.077 ampere).

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Review activities: Army -
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MS25327

MILITARY STANDARD

FED. SUP CLASS 5945

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			6	4			6	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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User activities: Army -
Navy -
Air Force -

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Procurement Specification: MIL-R-6106

P.A: USAF - 85
Other Cust: Navy - AS

International interest:

SUPERSEDES:

TITLE: RELAYS, ELECTROMAGNETIC, 10 AMPERE, 4 PDT, TYPE I, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25327

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Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Vibration - random	
Applicable specification	N/A
Power spectral density	N/A
RMS G min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude)

	80,000 ft
Coil to case	500 V rms
Aux contacts	500 V rms
All other points	500 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc, 60 amperes ac
Rupture current	50 amperes dc, 80 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C inspections are not applicable

(M) Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

FED. SUP CLASS 5945

APPROVED 12 May 1958 REVISED (M) . For changes see pages 3 and 6

U.S. Government Printing Office 1989-604-033/12246M

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